

A Consensus Statement on self-knowledge conceptualization, measurement, outcomes and changeability

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Abstract

Self-knowledge plays a central role in contemporary psychological science across various domains, including interpersonal relationships, moral behaviour and health. Despite its importance, many fundamental questions remain. We conducted a pre-registered, expert-based consensus process to address four key gaps in research on self-knowledge: its conceptualization, measurement, outcomes and changeability. Seventeen experts from diverse subfields of psychology participated in a structured Delphi process guided by four facilitators and an external advisor. The panel developed a consensus definition of self-knowledge as the extent to which a person has accurate perceptions of their own relatively stable characteristics and momentary states. Experts further agreed that self-knowledge is largely domain-specific, context-dependent in its benefits, and malleable in principle but difficult to change in practice. Measurement was identified as a central challenge, and avenues for refinement in future work were proposed. Consensus was weaker regarding the existence of a domain-general factor of self-knowledge and shared underlying processes across domains. Overall, the findings clarify where experts converge, where debates persist and what should be prioritized in future research, providing a crucial foundation for advancing the study of self-knowledge across fields.

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Introduction

The value of self-knowledge has been emphasized across cultures and eras – from the Upanishads in ancient India and Confucian writings in China to the Greek maxim ‘know thyself’, originating at the Oracle of Delphi. Across these traditions, understanding oneself is regarded as foundational to personal growth and a meaningful life. In modern times, self-knowledge continues to capture public interest, as evidenced by the widespread popularity of self-help books and blogs presenting self-knowledge as a cornerstone of self-improvement^{1–3}.

The concept of self-knowledge has also gained traction in psychological science and has become a unifying theme across multiple subfields⁴. Indeed, the relevance of self-knowledge spans numerous domains, including mental and physical health^{5–7}, occupational functioning^{8,9}, moral behaviour^{10,11} and interpersonal relationships^{12,13}. For instance, individuals must understand their emotional states, maladaptive patterns or unhealthy routines to maintain their physical and mental health; have insight into their interests and abilities to make informed career choices; and be aware of how they typically respond to others to navigate social interactions effectively¹⁴. Reflecting this wide-ranging impact, there is growing interest in interventions that enhance self-knowledge to foster socially valued outcomes, such as prosocial behaviour and psychological well-being^{15,16}.

Despite its broad relevance, the study of self-knowledge remains marked by unresolved foundational issues that have hindered theoretical integration and cumulative scientific progress. First, despite the widespread use of the concept, there is no broadly accepted definition of self-knowledge. Some definitions emphasize accurate insight into one’s own behaviour and characteristics as central to self-knowledge^{4,17,18}. Other conceptualizations emphasize the subjective experience of knowing oneself^{19,20}, as captured in concepts such as interoceptive awareness²¹, self-mentalizing²², felt authenticity²³ and subjective self-knowledge¹⁹. Critically, the distinction between the accuracy of self-perceptions versus the feeling of knowing is not always made explicit across the broader literature, which contributes to persistent conceptual and methodological ambiguities. In addition, it remains unclear whether self-knowledge is primarily domain-specific and varies across different characteristics (such as personality traits and abilities) or whether it reflects a more general tendency that spans multiple domains²⁴. Establishing agreement on a clear definition and clarifying whether self-knowledge is best conceptualized as a domain-specific or global construct have important implications for theory and measurement.

Second, measuring self-knowledge involves evaluating the convergence between individuals’ self-views and an external standard as a criterion. For example, a common operationalization of self-knowledge is self–other agreement^{25,26}, which compares how individuals perceive themselves with how others perceive them with regard to their behaviour or specific characteristics. However, the validity of this approach depends on the valid assessment of both perspectives and on selecting an appropriate convergence metric (for example, correlation, mean-level difference or profile similarity), which makes measuring self-knowledge more demanding than measuring other latent constructs, such as personality traits. Identifying a suitable criterion is often especially difficult because many measures only approximate an objective standard (such as when test performance is used as a proxy for actual ability). Reflecting this complexity, a wide range of measurement approaches have been proposed²⁶. However, little is known about the degree to which these methods converge or capture

the same underlying construct²⁷, which complicates comparison of findings across studies and domains²⁸.

Third, whereas some research suggests positive effects of greater self-knowledge on health and interpersonal functioning^{16,29,30}, other work points to potential downsides of accurate self-views. For example, being confronted with one’s shortcomings or undesirable characteristics might be associated with increased psychological distress³¹, leading to active avoidance of such information^{32,33}. Accordingly, some scholars have argued that mildly distorted but favourable self-views – so-called positive illusions – might be more beneficial for psychological functioning than full accuracy^{34–36}. These contrasting perspectives underscore that the outcomes associated with self-knowledge remain poorly understood, which raises the question of when greater self-knowledge is desirable and under what conditions it might have unintended or adverse consequences.

Fourth, the extent to which self-knowledge can change – and whether it can be changed intentionally – is particularly important if self-knowledge contributes to well-being and interpersonal functioning. Moreover, personality and behaviour change interventions often rely on individuals’ motivation and goal setting to initiate change^{37,38}. If self-knowledge helps shape people’s goals for changing their traits or behaviours – as some research suggests^{15,16,39} (but see ref. 40 for contradictory evidence) – then understanding the malleability of self-knowledge and the different ways through which changes in self-knowledge might occur could inform the design of effective interventions.

The issues described above underscore the need for rigorous, systematic and collaborative efforts to advance the study of self-knowledge. The value of collaborative efforts for advancing science in general^{41,42} – and psychological science in particular^{43,44} – is increasingly being recognized. In the case of self-knowledge, a collaborative approach is particularly valuable because many key questions are conceptual and methodological in nature, such as how self-knowledge should be defined and measured, respectively. Even empirically tractable questions (such as identifying the outcomes of self-knowledge) are difficult to synthesize due to variation in context, measurement and terminology. A structured expert-based approach can therefore help integrate diverse perspectives, clarify conceptual and methodological ambiguities, and identify areas of consensus and contention to guide future research.

In this Consensus Statement, we used the Delphi method⁴⁵ to synthesize expert perspectives on four key areas in the study of self-knowledge: its conceptualization, measurement, outcomes and changeability. These four areas are interdependent: conceptualizing self-knowledge is a prerequisite for developing valid measurement approaches, valid measurement is required to evaluate outcomes reliably, and understanding outcomes informs whether and how self-knowledge should be intentionally changed. By systematically integrating perspectives from different subfields, this consensus approach reduces fragmentation, mitigates jingle-jangle fallacies, and strengthens the conceptual and methodological foundation for the study of self-knowledge while preserving a diversity of viewpoints.

Methods

Transparency and open science

Study materials, data and analysis scripts are available in the Supplementary Information and Supplementary Data. The study design and analytic approach were pre-registered on the Open Science Framework (<https://osf.io/k82xv/>), with all deviations documented in Supplementary Note 1.

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Expert panel

The consensus-building process was guided by four facilitators (I.T., M.B., N.C. and A.S.). Experts were selected by the facilitators based on their recognized expertise in self-knowledge, with efforts to ensure a diverse panel in terms of gender, academic seniority, geographical location and psychological subfields. The facilitators conducted a literature review to identify leading scholars and contacted colleagues known to be working on self-knowledge. The selection was not intended to be exhaustive but aimed to capture a broad range of perspectives. However, all experts are affiliated with institutions in Western countries and perspectives from non-Western cultural contexts are therefore not represented.

A total of 17 experts (8 women, 9 men) were invited to participate and all agreed. The number of participants was guided by recommendations on the Delphi method⁴⁵ and prior studies that used similar consensus methods⁴⁶. Expert panel members ranged in age from 30 to 66 years (mean = 42.0, s.d. = 8.5). Most were full or associate professors ($n = 11$); the remainder were assistant professors and postdoctoral researchers. Experts could indicate more than one area of expertise; the majority reported primary expertise in personality ($n = 12$) and social psychology ($n = 9$), reflecting the centrality of these disciplines to the topic of self-knowledge. Additional areas represented by at least one expert included clinical, cognitive, environmental, experimental and quantitative psychology. At the time the Delphi process started, experts reported a median of 12 years of experience with the topic of self-knowledge (interquartile range = 6–16) and rated their own level of expertise as high (median = 4 on a 5-point scale from ‘very low’ to ‘very high’).

The facilitators also consulted with Professor Daniel Leising who has expertise in best practices for consensus-building processes⁴⁷.

Procedure

We conducted a structured consensus-building process based on the Delphi method⁴⁵ adapted for qualitative research questions^{46,48,49}. The Delphi method was used because it provides a structured, iterative and transparent approach to gather and synthesize expert judgement on complex, conceptual and methodological questions where empirical data are limited or heterogeneous. The process was structured to reflect the logical dependencies among issues: agreement on conceptualization was considered a prerequisite for meaningful discussion of measurement, outcomes and changeability.

The procedure closely followed guidelines for consensus-building processes⁴⁷ and unfolded over four stages (Fig. 1). All experts provided informed consent for each survey and agreed to openly share their responses, including open-ended comments (see Supplementary Data). Reporting of methods and results follows recommendations from the Conducting and REporting of DELphi Studies (CREDES) guidelines⁵⁰.

Stage 1: Gathering open-ended responses to derive initial themes.

Stage 1 consisted of an online survey (Supplementary Note 2). Experts first provided basic demographic information, along with details about their current position, primary areas of expertise and familiarity with the concept of self-knowledge. They then received detailed information about the goals of the study, before responding to five open-ended questions:

1. How would you define self-knowledge?
2. To what extent is self-knowledge domain-specific or global (i.e., domain-general)?
3. How do you think self-knowledge can best be measured?
4. Is more self-knowledge always better or can a person have too much self-knowledge? What would be appropriate criteria for testing whether it is ‘better’?
5. Can self-knowledge be increased? If so, how?

Experts were encouraged to express their own views and had the option to reference relevant literature. Responses were not limited in length. Experts received individualized survey links that enabled them to pause and return to revise their responses at any time before submission.

Stage 2: Assessing agreement with initial themes.

Stage 2 involved a follow-up survey (Supplementary Note 3) in which experts evaluated the 29 themes extracted from the open-ended responses in Stage 1. Themes were clustered under the five guiding questions (definition, specificity, measurement, outcomes and changeability). Experts rated their agreement with each theme on a 5-point Likert-type scale where 1 = strongly disagree, 2 = disagree, 3 = neither agree nor disagree, 4 = agree and 5 = strongly agree (note that the use of a 7-point Likert-type scale was pre-registered; however, a 5-point scale was mistakenly implemented in the Stage 2 survey). To inform their ratings, experts had access to all original Stage 1 responses – including their own – for each question (but without information about who wrote which response). Each survey page also included an open text box for optional comments or elaborations. Experts were invited to explain why they agreed or disagreed with particular themes and to suggest potentially overlooked themes. Finally, for each set of themes, experts were shown their original Stage 1 response and asked whether they wished to revise it.

Stage 3: Revising themes via in-person deliberation.

Stage 3 involved a 3-day, in-person meeting. All expert panel members were invited to the meeting, and nine were able to attend. The original goal of this meeting (per the pre-registration) was to focus on themes that had low consensus in Stage 2, but the facilitators ultimately decided to review and discuss all themes in detail. This change was prompted by comments in the Stage 2 survey indicating that some themes might have generated agreement due to vague wording (for example, “it may

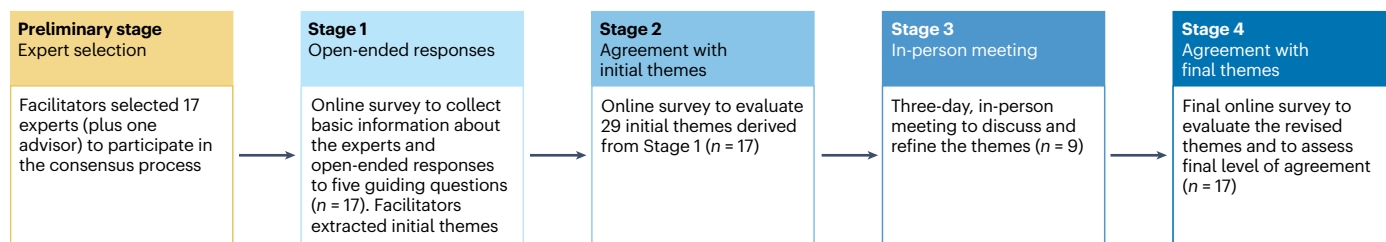


Fig. 1 | The consensus-building process. Flowchart summarizing the stages of the consensus process.

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be that ..." or "it can be that ...") and other themes combined multiple claims, thereby making endorsement more difficult. Experts received a printed packet with all Stage 1 responses to refer to during the discussions. The facilitators led structured discussions aimed at clarifying conceptual issues, refining or removing themes, and identifying areas of agreement and disagreement. To guard against groupthink and premature consensus, discussions were designed to be rigorous and inclusive, and facilitators avoided leading questions or framing themes in a way that might bias agreement. All discussions were audio-recorded for later reference. Themes were revised based on the discussions, and a tentative consensus definition of self-knowledge was developed.

Stage 4: Assessing agreement on final themes. Stage 4 involved a final online survey (Supplementary Note 4) in which all expert panel members were invited to participate, including those who had not attended the in-person meeting (Stage 3). Stage 4 provided an opportunity to confirm, revise and further clarify expert positions in light of the collective deliberations and refinements made throughout the process. Instructions emphasized that two of the original questions (concerning outcomes and changeability) had been slightly reworded to better reflect issues raised during the Stage 3 discussions. The revised questions read as follows: "Is more self-knowledge always beneficial or can a person have too much self-knowledge? What would be appropriate criteria for testing whether it is 'beneficial'?" and "Can self-knowledge change? If so, how?".

The revised themes were again clustered under the five guiding questions. Experts rated their agreement with each theme using a 7-point Likert-type scale (1 = strongly disagree, 2 = disagree, 3 = somewhat disagree, 4 = neither agree nor disagree, 5 = somewhat agree, 6 = agree and 7 = strongly agree). For each theme, if an expert panel member responded with a rating of ≤ 5 , they were prompted to briefly explain why they do not (fully) agree with this theme. For the question on the definition of self-knowledge, the survey presented the proposed consensus definition developed in Stage 3. Experts rated their agreement with this definition using the same 7-point scale and were asked to specify which part or parts of the definition they disagreed with, if any, and why. The final responses in Stage 4 served as the basis for synthesis.

Data analysis

Open-ended responses collected in Stage 1 were analysed using an inductive content analysis approach⁵¹. Each of the four facilitators independently reviewed all responses to identify recurring themes. ChatGPT (version GPT 4o, June 2024) was used as an additional tool to support consistency and comprehensiveness in theme extraction. The model was provided with the dataset and prompted to identify recurring topics using a content analysis approach (see Supplementary Note 5 for the exact prompt). Facilitators then convened to compare results, resolve discrepancies and agree on the set of themes to be rated in Stage 2.

Ratings from Stages 2 and 4 were analysed descriptively, focusing on the level of agreement among experts. Following prior research⁴⁸, strong consensus was considered reached when at least 70% of experts either agreed or strongly agreed with a theme (selecting a rating of 6 or 7 on the 7-point scale), or disagreed or strongly disagreed (selecting a rating of 1 or 2). Broad consensus was considered reached when at least 70% of experts expressed at least slight agreement (selecting a rating of 5, 6 or 7) or slight disagreement (selecting a rating of 1, 2 or 3).

There is no universally accepted threshold for defining consensus. Stricter cut-offs (75% or 80%) are commonly applied in some research areas⁵⁰ and more lenient cut-offs (starting from 60%) are regularly

used in others⁵². We therefore adopted a 70% threshold consistent with prior work in (social) psychological research⁴⁸, but acknowledge that any cut-off is ultimately somewhat arbitrary. Importantly, our substantive conclusions would remain unchanged if a stricter threshold (for example, 75%) was applied.

Due to the inadvertent use of a 5-point scale in Stage 2, strong consensus at that stage reflects that at least 70% of experts strongly agreed (rating of 5) whereas broad consensus reflects that at least 70% of experts agreed or strongly agreed (rating of 4 or 5). Compared with the 7-point variant originally planned, the 5-point scale is slightly more conservative and yields broad and strong consensus somewhat less frequently (40% and 20% of options, respectively, versus 43% and 29% with the 7-point scale).

Results

For concision, we only report the final themes endorsed in Stage 4 here, together with corresponding levels of consensus (Table 1 and Fig. 2). Details on how initial themes were derived and revised in earlier stages of the process are provided in Supplementary Note 6 and Supplementary Table 1.

Overall, experts reached strong consensus for 15 of the 17 final themes (including the definition of self-knowledge) and broad consensus for the remaining 2 themes. Agreement was highest for claims about measurement and lowest for claims about specificity. Results for each of the five guiding questions are summarised below along with the main considerations that qualified expert consensus.

Definition

Experts collaboratively developed a unifying definition of self-knowledge during the in-person expert meeting (Stage 3) based on the themes derived from the Stage 1 responses, which subsequently reached strong consensus in the final survey (Stage 4): "Self-knowledge describes the extent to which a person has accurate perceptions of their own relatively stable characteristics and momentary states (e.g., personality features, abilities, affect, motivations, preferences, physical appearance, behaviors)."

Specificity

There was strong consensus for the idea that self-knowledge is primarily domain-specific (Theme S1), which was based on empirical evidence (for example, variation in self-knowledge across ability domains^{53–55}) as well as conceptual reasoning. By contrast, consensus was only broad for the two related themes: that self-knowledge could also be defined at a global level, analogous to the general (*g*) factor of intelligence (Theme S2); and that certain processes and mechanisms (for example, biases or motives) might underlie self-knowledge across domains (Theme S3).

In open-ended comments, experts questioned whether there is sufficient empirical evidence for a domain-general component of self-knowledge. Several experts noted that although the analogy to the *g* factor of intelligence (Theme S2) is conceptually appealing, it might be overly simplistic and empirically premature. Other experts highlighted that the proposed shared processes (Theme S3) might not operate uniformly across all domains, suggesting instead that different mechanisms could govern self-knowledge in distinct contexts. For example, self-knowledge about internal characteristics (such as emotional experience) might rely on one's ability to perceive and interpret inner feelings, whereas self-knowledge about more observable characteristics (such as skills or behavioural tendencies) might rely on accurately processing external feedback.

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Table 1 | Themes and corresponding consensus ratings from the final stage of the consensus process

Theme	Agreement (%)		Mean (s.d.)
	Broad consensus	Strong consensus	
Definition			
Self-knowledge describes the extent to which a person has accurate perceptions of their own relatively stable characteristics and momentary states (e.g., personality features, abilities, affect, motivations, preferences, physical appearance, behaviors)	100	88.2	6.59 (0.71)
Specificity (S)			
S1. Self-knowledge in one domain does not necessarily generalize to other domains	100	100	6.82 (0.39)
S2. Even if self-knowledge is domain-specific to a certain extent, it can be defined at a global level, analogous to the general (g) factor of intelligence	88.2	35.3	5.47 (1.18)
S3. There are certain processes and mechanisms (e.g., biases, motives) that underlie self-knowledge across domains	100	58.8	6.06 (0.97)
Measurement (M)			
M1. The measurement of self-knowledge should involve a comparison of self-perceptions with a relevant criterion	94.1	94.1	6.65 (0.79)
M2. The measurement of self-knowledge is challenging because it involves the assessment of two separate constructs (i.e., a self-perceived characteristic and a relevant criterion), plus an index of comparison among the two	94.1	82.4	6.35 (1.32)
M3. What constitutes a relevant criterion depends on the domain and type of self-knowledge one is interested in, but it should match the characteristic in question (e.g., in terms of specificity)	100	94.1	6.76 (0.56)
M4. Some characteristics are directly observable (e.g., height) and can be measured straightforwardly, whereas other characteristics are latent (e.g., personality traits) and require proxies for measurement, thus posing additional challenges for the measurement of self-knowledge	100	100	6.94 (0.24)
Outcomes (O)			
O1. More self-knowledge is not always beneficial	94.1	70.6	6.18 (1.01)
O2. Criteria to assess whether self-knowledge is beneficial include, but are not limited to, the level of intrapersonal functioning (e.g., health, subjective well-being), the level of interpersonal functioning (e.g., quality of social relationships), and goal achievement	100	82.4	6.47 (0.80)
Changeability (C)			
C1. Self-knowledge is malleable	100	76.5	6.41 (0.87)
C2. In most domains, increasing self-knowledge is difficult	88.2	82.4	5.94 (1.09)
C3a. Changes in self-knowledge can occur through obtaining new information about the self	100	94.1	6.59 (0.62)
C3b. Changes in self-knowledge can occur through obtaining new information about others	100	88.2	6.41 (0.71)
C3c. Changes in self-knowledge can occur through using* existing information about the self and/or others in a different way (e.g., integrating, interpreting, accessing)	100	94.1	6.59 (0.62)
C3d. Changes in self-knowledge can occur through changing one's level of the characteristic in question	88.2	70.6	6.00 (1.22)
C4. Self-knowledge can be changed both intentionally (e.g., through self-experimentation) and incidentally (e.g., through experiences)	100	94.1	6.65 (0.61)

Note. Themes are presented verbatim and were rated in Stage 4 of the consensus process on a 7-point scale ranging from 1=strongly disagree to 7=strongly agree. Percentages indicate the proportion of expert panel members whose responses met the defined threshold for consensus. Strong consensus was defined as $\geq 70\%$ of expert panel members indicating agreement (6) or strong agreement (7). Broad consensus was defined as $\geq 70\%$ of expert panel members indicating slight agreement (5), agreement (6) or strong agreement (7).

Measurement

There was strong consensus for all four final themes related to measuring self-knowledge. Experts agreed that that “the measurement of self-knowledge should involve a comparison of self-perceptions with a relevant criterion” (Theme M1), while also emphasizing that this task is inherently complex because it requires assessing two separate constructs (the self-perception and the criterion) plus an index of their correspondence (Theme M2). There was also consensus that the relevance of a given criterion for measuring self-knowledge depends on the domain and characteristic in question (Theme M3). For example, it might be inappropriate to use a highly specific behaviour in a single situation (for example, arriving late to a meeting) as a benchmark for assessing self-knowledge of a broad trait such as conscientiousness

because such behaviour can be influenced by situational factors unrelated to the trait (for example, public transportation delays; for similar reasoning, see refs. 56–58). Finally, experts agreed that “some characteristics are directly observable (e.g., height) and can be measured straightforwardly, whereas other characteristics are latent (e.g., personality traits) and require proxies for measurement, thus posing additional challenges for the measurement of self-knowledge” (Theme M4).

Outcomes

Both outcome-related themes received strong consensus in Stage 4. Experts endorsed the view that the benefits of greater self-knowledge are conditional rather than universally positive (Theme O1: “More self-knowledge is not always beneficial”), referring to mixed

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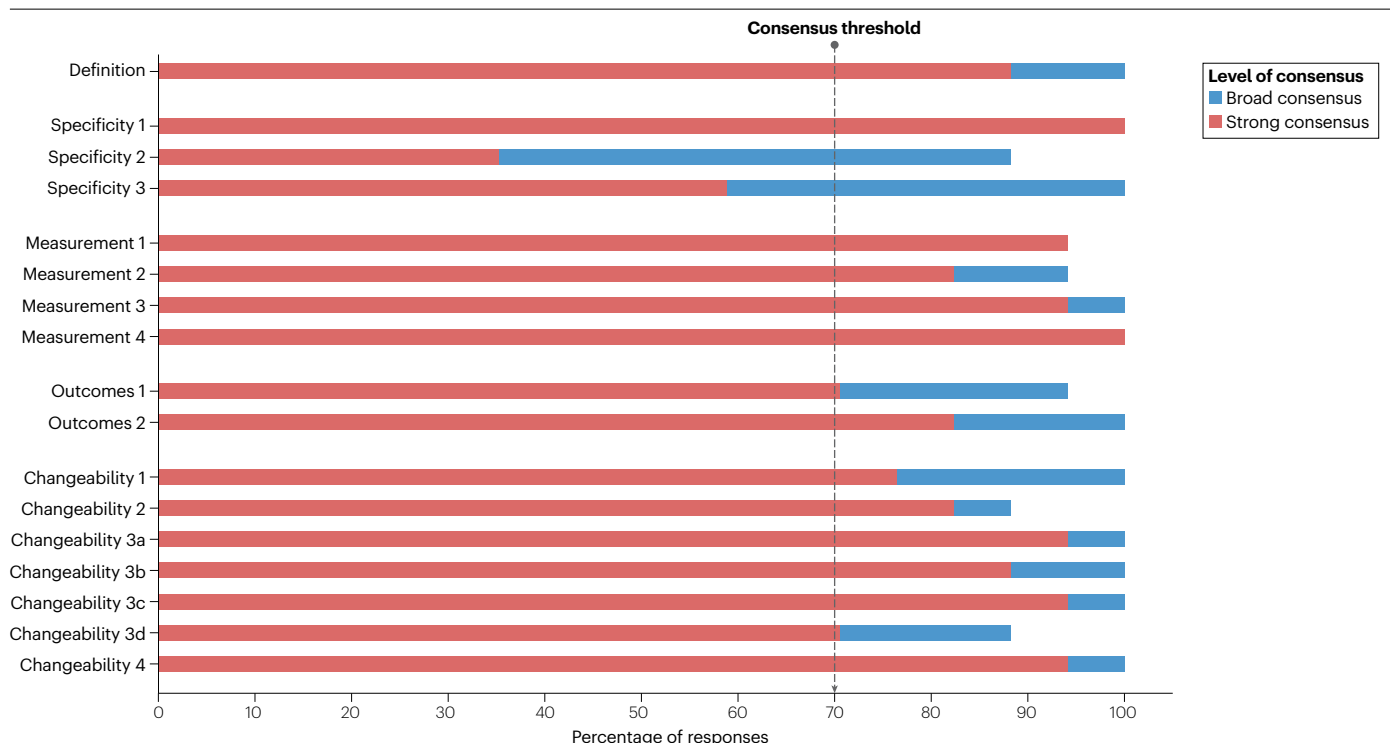


Fig. 2 | Results of the consensus-building process. The degree of consensus with the identified themes in the final stage of the consensus process. Strong consensus (red) was defined as $\geq 70\%$ of expert panel members indicating agreement or strong agreement. Broad consensus (blue) was defined as $\geq 70\%$

of expert panel members indicating slight agreement, agreement or strong agreement. See Supplementary Note 7 for the script that replicates the results and figure.

empirical findings. Specifically, experts noted that some studies link self-knowledge to positive outcomes^{13,59,60}, whereas other studies show advantages of self-enhancement in certain domains^{28,61}, and still others find null effects⁶². Experts also agreed on three broad outcome domains inspired by prior research (Theme O2): intrapersonal functioning (for example, health and subjective well-being), interpersonal functioning (for example, the quality of social relationships) and goal achievement.

That said, open-ended comments revealed concerns about the clarity and formulation of the two outcome-related themes. Some experts noted that Theme O1 could be worded more precisely; others questioned the separation of goal achievement from intrapersonal and interpersonal functioning in Theme O2. As discussed in Stage 3, one rationale for this distinction was the conceptual difference between goal achievement, which typically refers to the pursuit and attainment of outcomes in domains such as education or work, and outcomes related to intrapersonal or interpersonal functioning, which centre on a person's subjective experience or social dynamics, respectively.

Changeability

There was strong consensus on all final themes regarding the changeability of self-knowledge. Experts agreed that self-knowledge is malleable (Theme C1), although improving it is difficult in most domains (Theme C2). These judgements were informed primarily by theoretical considerations and expert intuition, and in open-ended comments experts noted their caution in strongly endorsing these claims given the scarcity of direct empirical evidence. Experts further endorsed multiple routes through which changes in self-knowledge could occur, including

intentional efforts (such as self-experimentation or therapy), incidental experience (such as life events) and informational pathways (such as feedback, reflection or social comparison) (Themes C3a–C3c). However, there were some reservations regarding Theme C3d – that changes in self-knowledge can occur through changes in the characteristic itself – because this perspective might conflate a change in the underlying trait or state with a change in how accurately the person perceives their level on that trait. Finally, experts strongly agreed that self-knowledge can change both intentionally and incidentally (Theme C4), highlighting the multiple pathways through which self-knowledge might develop.

Discussion

Our consensus process based on the Delphi method produced a shared definition of self-knowledge and clarified key areas of agreement and disagreement among experts. Here we discuss additional considerations, challenges and ways forward drawn from discussions during the process and the open-ended comments for each guiding question. Together, these insights map the current state of the field, highlight clear directions for future research (Table 2) and inform practical recommendations for conducting research on self-knowledge (Box 1).

Defining self-knowledge

The consensus process yielded a shared definition of self-knowledge (see Table 1) that distinguishes self-knowledge from self-beliefs. Specifically, self-knowledge is defined by the degree to which perceptions of one's relatively stable characteristics and momentary states are accurate and therefore correspond to actual characteristics or

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behaviours, whereas self-beliefs refer to a person's subjective views about themselves that might or might not be accurate.

A key epistemological challenge is specifying what exactly constitutes 'accuracy' in this context. 'Accuracy' implies the existence of a reference point – that is, an external standard against which self-perceptions can be evaluated. However, in psychological research such standards are rarely definitive and can typically only be approximated. Latent constructs such as personality traits, preferences or motivations, in particular, lack definitive external standards and do not have a single gold-standard criterion. Researchers must therefore often rely on imperfect yet informative proxies, such as informant reports or behavioural indicators. Given these challenges, we recommend that researchers clearly justify their choice of reference points or external criteria for establishing accuracy and spell out any limitations to those criteria.

It is also important to note that the consensus definition defines self-knowledge in terms of perception. Perception, including self-perception, is shaped by both bottom-up and top-down processes^{63,64} and is therefore inherently constructive and flexible. Thus, the consensus definition implies that self-knowledge (similar to other forms of perception) is not static but, rather, dynamically formed and updated. Accordingly, the consensus definition inherently captures that self-knowledge might vary within a person and across contexts based on situational and motivational influences.

Domain specificity of self-knowledge

Experts strongly agreed that self-knowledge is best conceptualized as domain-specific. For instance, alexithymia is commonly understood as involving a selective deficit in emotional self-knowledge, rather than a global deficit in self-insight⁶⁵. Moreover, the cues available for forming accurate self-views differ by domain: whereas self-knowledge about cognitive abilities might be informed by relatively objective feedback (such as formal testing), self-knowledge about emotional states might rely on early socialization experiences (such as whether one's emotions were acknowledged, named or dismissed by caregivers)⁶⁶. In line with

these ideas, evidence on self-knowledge regarding different abilities points to domain specificity^{53,54,67}. Thus, a person might have high self-knowledge in one domain and low self-knowledge in another.

At the same time, experts were open to the possibility of a broader, domain-general component, although the themes suggesting domain-general (global) aspects of self-knowledge received the lowest consensus in the entire process. This perspective acknowledges that certain cognitive and motivational processes might shape self-knowledge across domains. These mechanisms could operate by influencing how individuals attend to, interpret or selectively encode information about themselves²⁴, or by motivating consistent patterns of attributing positive characteristics to oneself in multiple domains (self-enhancement or halo bias^{68,69}). Empirically, however, open questions remain, such as to what extent self-knowledge across different domains converges, how shared versus domain-specific mechanisms operate and whether a general core (comparable with the *g* factor of intelligence) can be meaningfully identified. This debate parallels the literature on emotional intelligence, where claims of a global factor have been met with scepticism based on increasing evidence for domain-specific skills^{70,71}. Likewise, there is little agreement on which domains would meaningfully constitute 'global' self-knowledge in the first place.

A key task for future research will therefore be to determine whether and how self-knowledge converges across domains. Promising approaches include assessing self-knowledge across multiple domains within the same study to formally evaluate the psychometric validity of general versus domain-specific models and investigating whether shared cognitive or motivational processes (such as self-enhancement or attentional biases) account for cross-domain correlations.

Measuring self-knowledge

Consensus was strongest for measurement-related themes overall. Experts agreed that assessing self-knowledge is inherently difficult and requires careful methodological choices to draw valid inferences about the accuracy of self-perceptions. A core challenge is that self-knowledge cannot be measured directly, for example, by simply asking people

Table 2 | Future research directions

Area	Future direction	Open questions
Conceptualization	Conceptualize the structure and breadth of self-knowledge	Does self-knowledge converge across domains or is it domain-specific? What cognitive or motivational processes explain cross-domain correlations, if they are meaningful? What are additional ways in which a person can know themselves beyond knowledge of broad, relatively stable traits? How can self-knowledge be situated within a broader conceptual and theoretical framework alongside related constructs such as self-concept, self-representation and self-beliefs?
Measurement	Measure complex and global forms of self-knowledge	How can complex forms of self-knowledge (such as how elaborated a person's self-knowledge is) be measured? Which domains should be captured in a measure of 'global' self-knowledge, if this construct is meaningful? What alternative data sources (such as behavioural traces or autobiographical materials) can provide insights into self-knowledge that complement traditional convergence measures?
Outcomes	Clarify the outcomes of self-knowledge	How does self-knowledge influence outcomes across life domains? For whom and under what conditions is self-knowledge desirable? To what extent do the outcomes of self-knowledge vary as a function of the level of the underlying trait? What are the distinct effects of rank-order accuracy versus mean-level bias, and of absolute versus idiographic accuracy? By what mechanisms does self-knowledge translate into outcomes?
Changeability	Elucidate how self-knowledge develops and changes	How does self-knowledge develop across the lifespan? Do short-term changes in self-knowledge consolidate into long-term developmental shifts? What cognitive and motivational processes underlie self-knowledge change? How do intentional versus incidental pathways of change compare?

Box 1 | Recommendations for research on self-knowledge

Conceptualization

- Distinguish self-knowledge from how knowledgeable one subjectively feels
- Specify which domain of self-knowledge is under investigation (for example, traits or behavioural variability)
- Specify which accuracy concept is of interest (for example, rank-order versus absolute accuracy)

Measurement

- Select the assessment approach that best fits the research question and context
- Use highly reliable instruments for both self-report and external criteria
- When using informant reports as the external criterion, triangulate across multiple sources who observe the target in different contexts (for example, family, peers and co-workers)
- When using trait labels (for example, adjectives such as 'odd' or 'honest'), provide raters with clear construct definitions and behavioural indicators
- Specify standardized reference groups (for example, 'compared to most people' or 'compared to peers of the same age and gender') to reduce bias from raters using different frames of reference

- Explicitly justify methodological decisions, acknowledge assumptions and trade-offs, and be transparent about limitations of the approach chosen

Outcomes

- Frame research questions in terms of for whom, when and in which domains self-knowledge is beneficial, rather than whether self-knowledge is generally 'good' or 'bad'
- Incorporate multiple data sources to capture outcomes, especially in interpersonal and societal domains (for example, partners' perspectives and behavioural indicators)
- Ensure that the domain of self-knowledge under scrutiny aligns with the outcome of interest (for example, cognitive self-knowledge for professional performance and athletic self-knowledge for sports outcomes)
- Use analytic methods that are well suited for testing congruence hypotheses (such as response surface analysis)

Changeability

- Prioritize longitudinal designs to track both short-term and long-term changes

how much they 'know themselves'. Instead, self-knowledge is typically inferred from the degree of correspondence between a person's self-views and an external criterion, both of which are prone to measurement error. Moreover, combining two error-prone measures might reduce reliability and introduce systematic biases – a concern well documented in the literature on difference scores^{72,73}. To mitigate this issue, we recommend using highly reliable instruments for both components.

A central question is what constitutes a suitable criterion – that is, how do researchers establish what people are actually like? The answer depends on the domain and aspect of self-knowledge under investigation and the validity of available indicators. Performance tests might provide appropriate benchmarks for cognitive abilities⁶⁷, whereas (aggregated) informant reports might be more informative for socially visible attributes such as humour⁷⁴. For less visible traits such as neuroticism, self-reports of subjective feelings or observable behaviour in stressful situations might be most suitable⁷⁵. However, in some domains the relevant characteristics are so internal (for example, personal motives or intentions) that no independent criterion exists. In these cases, self-reports might offer the only practical access to the construct, even though they cannot serve as an external benchmark for evaluating self-knowledge.

Indirect measures (such as the implicit association test⁷⁶, the picture story exercise⁷⁷ or physiological indicators of affective reactivity⁷⁸) might serve as useful benchmarks when researchers seek to compare explicit self-perceptions with less conscious, implicit assessments of the same attribute^{26,79,80}. However, caution is warranted in the use and interpretation of implicit measures^{81,82}.

Repeated assessments of actual behaviour across contexts (using methods such as the electronically activated recorder⁸³ or experience sampling) might be necessary to capture moment-to-moment fluctuations in trait enactment and thereby evaluate self-knowledge about behavioural variability^{84,85}. Finally, for questions of meta-accuracy

(knowledge of how others perceive oneself), the relevant criterion is informant reports that reveal how others actually perceive oneself⁸⁶.

The choice of a criterion is further complicated by the nature of the characteristic in question. Some attributes (such as height) are directly observable and objectively quantifiable. Others (such as personality traits or motivations) are latent and must be assessed through proxies such as informant reports or behavioural indicators. When imperfect criterion measures are used, discrepancies between self-ratings and criterion measures are subject to several alternative interpretations. For example, discrepancies between self-ratings and informant ratings might reflect limited insight on the part of the individual or limited knowledge on the part of the informant, or both. Informant reports are also often positively biased when informants like the target person they evaluate^{87,88}. Ideally, informant data should be triangulated across multiple sources who have different relationships with the target and know them in different contexts.

Even with an appropriate criterion, challenges regarding the convergence metric might arise. First, different research questions call for different metrics. Difference scores (for example, the discrepancy between self-rated and actual test performance) capture over-estimation or under-estimation; between-person correlations assess knowledge of one's standing relative to other people on a particular characteristic; and profile correlations assess knowledge of which characteristics are higher or lower within oneself (across a large set of characteristics).

Second, high convergence between two measures does not necessarily reflect high self-knowledge. For example, correlations between self-reports and informant reports might be inflated by shared bias: both raters might systematically overestimate or under-estimate a trait in the same direction, producing apparent agreement despite low actual accuracy⁸⁹. Conversely, divergence does not necessarily reflect poor self-knowledge. For instance, divergence between two raters

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(for example, self and informant) might stem from differences in how raters conceptualize an attribute and what they consider valid indicators of it: raters might interpret the label 'talkative' more consistently than 'odd', leading to greater self–other agreement on how 'talkative' a person is than on how 'odd' they are⁹⁰. In addition, because individuals usually form person judgements relative to others, divergence might arise from mismatched frames of reference. For example, students in selective academic environments might under-estimate their abilities because they compare themselves with unusually capable peers (the big fish–little pond effect^{91,92}). To mitigate such issues, we recommend that researchers provide raters with clear construct definitions and explicitly specify standardized reference groups (for example, 'compared to most people'⁹³ or 'compared to a representative sample from Germany'¹⁵).

Experts also noted that self-knowledge might go beyond the convergence between a self-view and a criterion for a specific trait or class of traits. To know someone well, it is important to have a nuanced, elaborated and contextualized portrait of their personality – how they function in time, place and role – rather than knowledge of broad traits only⁹⁴. However, contextualized aspects of self-knowledge – such as understanding one's motivations, triggers or reasons behind thoughts, feelings and behaviours – remain largely overlooked in existing measures of self-knowledge. Addressing this gap is an important task for future research. Researchers should also consider alternative sources of data to draw inferences about self-knowledge, such as behavioural traces (for example, life decisions or autobiographical materials). These indicators can reveal how people typically think, feel or act in real-life contexts, thereby providing insights into self-knowledge that could complement traditional convergence metrics.

Taken together, these reflections highlight that although accuracy is central to the construct of self-knowledge, it can take multiple forms and requires careful operationalization. Choices of self-report measure, criterion and indicator of convergence must be tailored to the research question, the type of characteristic under scrutiny and the availability of external criteria. Because little evidence exists on which indicators best capture different aspects of self-knowledge, we do not provide prescriptive recommendations about specific measures or methods. However, researchers should be explicit about their methodological decisions, acknowledge assumptions and trade-offs, and be transparent about limitations of the approach chosen. Such practices will enhance interpretability, facilitate meaningful comparisons across studies, and promote methodological innovation and cumulative science.

Outcomes of self-knowledge

Greater self-knowledge has been associated with higher relationship satisfaction, better health and better-informed life decisions^{13,59,60,95}. At the same time, high self-knowledge can present painful truths, potentially undermining self-esteem and subjective well-being^{28,31,34}. Reflecting this complexity, there was strong consensus that more self-knowledge is not always beneficial.

However, a key insight from the expert discussions was that the broad question of whether more self-knowledge is 'good' or 'bad' might be misguided. Rather than seeking a universal answer, researchers should ask whether self-knowledge is beneficial in a certain domain, under certain circumstances or for certain people. Moreover, the outcomes of self-knowledge might differ depending on whether one looks at intrapersonal functioning (such as subjective well-being), interpersonal functioning (such as relationship quality), goal attainment or broader social consequences (such as prosocial behaviour).

For example, recognizing one's shortcomings might challenge self-esteem in the short term but strengthen relationships in the long term by fostering humility and openness to feedback.

Moreover, various factors could moderate the outcomes of self-knowledge. People need to act on the information they have, so knowledge alone is unlikely to guarantee positive outcomes. For example, knowing one's strengths might not translate into achieving goals if anxiety prevents their use, and recognizing interpersonal shortcomings could prompt avoidance rather than efforts to improve social skills or communication. Future research should therefore investigate individual-level factors that shape who applies – or fails to apply – their self-knowledge in what way. Relatedly, more work is needed on the potential moderating role of trait levels: for example, knowing that one is high on intelligence might boost self-esteem, whereas knowing that one is low on intelligence might reduce it. In addition, researchers should consider that rank-order accuracy and mean-level bias can be independent and might be associated with distinct outcomes. That is, one could be biased in absolute trait levels yet correctly identify relative strengths and weaknesses.

The expert panel also highlighted that it is essential to consider multiple data sources to operationalize outcomes, especially in the interpersonal and societal domains where others' perspectives or behavioural indicators might provide crucial information. Relying exclusively on self-reports – as is common practice in several psychological subfields⁹⁶ – to measure outcomes poses two challenges: it assumes that individuals have accurate insight into the very outcomes being studied (for example, relationship quality) and it risks inflating associations with measures of self-knowledge because both predictor and outcome stem from the same source (the self). However, self-reports might remain indispensable for some outcomes (such as subjective well-being) because they capture experiences that cannot easily be observed externally.

In addition, investigating the outcomes of self-knowledge presents statistical challenges because the central question involves a congruence hypothesis: whether alignment between self-views and an external criterion predicts meaningful outcomes. A common approach to testing such hypotheses is difference scores that compare self-views and external criteria, which are then correlated with corresponding outcomes⁹⁷. However, this approach is problematic because collapsing self-views and external criteria into a single score loses information about the absolute levels of each variable, making it impossible to disentangle the effects of the trait itself from the effects of alignment. For example, because extraversion is positively related to well-being⁹⁸, if individuals who are more extraverted also have more self-knowledge of their extraversion, any apparent benefit of self-knowledge could simply reflect the main effect of extraversion. Consequently, difference scores can produce biased results^{27,28,99}.

A more rigorous alternative is response surface analysis, which enables researchers to test congruence hypotheses directly while disentangling main effects from alignment effects, thereby avoiding interpretational pitfalls of other approaches^{61,100} (but see ref. 27 for a more critical assessment of the advantages of response surface analysis). Critically, response surface analysis requires testing non-linear terms, including interactions, which tend to be small and difficult to detect^{101,102}. As a result, response surface analysis typically requires relatively large sample sizes. As a rule of thumb, sample sizes should at least be two to three times larger than those required to detect linear main effects¹⁰⁰. However, the sample size required for response surface analysis depends on study-specific design features, including

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the complexity of the congruence hypothesis under investigation. At present, no tools are available that enable precise sample size planning for response surface analysis¹⁰³.

The expert panel also noted that relative knowledge of one's strengths and weaknesses might sometimes matter more for outcomes than absolute accuracy on specific attributes. In other words, understanding which qualities distinguish oneself within one's own profile – what one is comparatively strong or weak at – might be more consequential than knowing one's exact standing on any single characteristic. This idiographic perspective could be particularly relevant for decision-making and goal setting. Future research should test whether idiographic self-knowledge predicts outcomes beyond the effects of absolute accuracy.

In general, experts cautioned against normative prescriptions about self-knowledge. Although self-knowledge is often regarded as valuable, there might be contexts in which having less self-knowledge serves the individual better – for instance, by preserving a positive self-image and increasing motivation. Similarly, in close relationships strict accuracy might be less important than achieving a 'shared reality' with one's partner^{104,105}. For example, imagine a couple in which both partners believe that Partner A is very organized and Partner B is very spontaneous. Even if these beliefs are somewhat exaggerated or objectively inaccurate, the alignment of their perceptions might enable them to coordinate roles and responsibilities smoothly, thereby facilitating predictability and relationship satisfaction.

Finally, some experts noted that self-knowledge might be valued as an end in itself. This idea resonates with long-standing philosophical perspectives (for instance, Socrates' claim that "the unexamined life is not worth living") that emphasize the intrinsic worth of striving for self-knowledge.

Together, these insights highlight the need for greater nuance and contextual sensitivity when studying the outcomes of self-knowledge. Future research should clarify the processes through which self-knowledge produces beneficial or detrimental outcomes and identify the factors that determine for whom and in what contexts self-knowledge matters most.

Changeability of self-knowledge

There was strong consensus on themes related to the changeability of self-knowledge. Most fundamentally, experts agreed that self-knowledge is malleable. This view was mostly grounded in theoretical considerations and expert intuition, but also aligns with emerging empirical evidence. For example, studies suggest that individuals might adapt their self-views in response to feedback^{15,39,106,107}. However, little is known about whether such short-term shifts consolidate into enduring change. Addressing this gap requires longitudinal research.

It is also unknown how self-knowledge develops across the lifespan. One possibility is that self-knowledge follows a maturational trajectory – akin to personality change^{108,109} – such that older adults have greater self-knowledge than children and younger adults due to accumulated experience. Alternatively, ageing might foster more rigid self-views and reduce opportunities for feedback, especially for individuals in high-status roles who might receive less candid input from lower-status others. Moreover, age-related cognitive changes might alter how self-relevant information is processed. Fuzzy trace theory suggests that adults increasingly rely on gist processing (prioritizing broad themes over precise details) whereas children rely more on verbatim processing^{110,111}. Such age-related shifts might complicate how self-related information is encoded and recalled, which

might undermine self-knowledge. Overall, clarifying how life experiences, social feedback and cognitive processing shape changes in self-knowledge is a key direction for future research.

Despite the likely malleability of self-knowledge, experts agreed that increasing self-knowledge might often be difficult. Arguably, increasing self-knowledge is especially difficult when self-views are inflated because individuals are motivated to preserve coherent, positive self-views – a tendency reflected in well-known phenomena such as self-enhancement^{69,112} and self-verification¹¹³. Consequently, new information about oneself might be selectively dismissed or insufficiently integrated into one's self-concept. From a cognitive perspective, self-knowledge can be viewed as a form of Bayesian belief updating, in which the self-concept acts as a prior theory about the self¹¹⁴. If that prior is held with high certainty – that is, a person is highly confident in their self-view – then even substantial new evidence might have little impact. These motivational and cognitive barriers align with the realistic accuracy model¹¹⁵, which posits that self-knowledge requires relevant behaviour to occur, become observable, be detected and, ultimately, be used. Failures at any of these stages (for example, due to defensiveness, inattention or limited information) can prevent change. Thus, even when opportunities for increasing self-knowledge arise, they might not lead to updated self-views unless individuals are open and able to process the corresponding information^{116–118}.

Experts identified several pathways through which self-knowledge might change, such as acquiring new information about the self (for example, through feedback or reflection^{119,120}), learning about others (which might shift one's frame of reference) and interpreting or integrating existing self-related information in new ways (for example, through mindfulness¹²¹). Although we discuss these pathways under the lens of change, they also reflect broader sources and processes that contribute to self-knowledge more generally. In other words, the inputs that enable self-knowledge to change could be the same ones through which self-knowledge initially forms and develops.

Experts also emphasized nuanced forms of change, including not just greater accuracy but also increased elaboration – that is, deeper and more differentiated understanding of one's own characteristics and tendencies⁹⁴. Increased elaboration might involve recognizing situational contingencies, typical emotional responses or how one's behaviour fluctuates across contexts^{84,122}.

Changes in self-knowledge might occur both intentionally and incidentally. Intentional change might be fostered through structured feedback, guided self-reflection or self-experimentation (for example, deliberately entering new situations to observe oneself). By contrast, incidental change might result from life events (for example, starting a new job, experiencing failure, travelling or becoming a parent) that gradually prompt revisions in self-perception, even in the absence of deliberate efforts¹²³. Given this range of possible mechanisms, future research should systematically compare different pathways with self-knowledge change and examine for whom and under which circumstances they take effect.

Another open question concerns the psychological processes underlying changes in self-knowledge. According to the TESSERA (triggering situations, expectancy, states/state expressions and reactions) model¹²⁴, enduring personality change emerges through repeated state-level changes (short-term shifts in thoughts, feelings and behaviours) that accumulate over time. Such repeated, momentary shifts in self-perception could arise both from intentional strategies and from general life experiences. Charting these dynamic processes

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Conceptualization 1 definition	Specificity 3 themes	Measurement 4 themes	Outcomes 2 themes	Changeability 4 themes
Consensus definition of self-knowledge: “Self-knowledge describes the extent to which a person has accurate perceptions of their own relatively stable characteristics and momentary states (e.g., personality features, abilities, affect, motivations, preferences, physical appearance, behaviors).”	Expert opinion Self-knowledge is primarily domain-specific	Self-perceptions should be compared with a matching criterion	Self-knowledge is not always beneficial for intra/interpersonal functioning and goal achievement	Self-knowledge can change both incidentally and intentionally
	Future directions Conceptualizing the structure and breadth of self-knowledge	Measuring complex and global forms of self-knowledge	Clarifying the outcomes of self-knowledge	Understanding how self-knowledge develops and changes
	Recommendations • Distinguish self-knowledge and perceived self-knowledge • Specify the domain of self-knowledge and the accuracy metric	• Match method to purpose • Ensure reliability • Triangulate informant data • Clarify constructs • Standardize comparison groups • Be transparent	• Contextualize benefits • Use multiple data sources and suitable congruence metrics • Align domain and outcome	• Prioritize longitudinal designs

Fig. 3 | Summary of key conclusions from the consensus process. The consensus process led to a consensus definition of ‘self-knowledge’. For each of the other guiding questions (specificity, measurement, outcomes

and changeability) expert opinion further gave rise to priorities for future research and recommendations.

and identifying the conditions that promote self-knowledge change represent key opportunities for future research.

Conclusion

This consensus effort highlights both the promise and the complexity of self-knowledge as a psychological construct. The expert panel converged on a shared definition and affirmed the importance of studying self-knowledge across domains, methods and outcomes, while also recognizing that the construct resists simple answers. Self-knowledge is conceptually multifaceted, shaped by cognitive and motivational processes, and requires careful methodological choices to capture with precision. Its outcomes vary by context and domain, and its development – whether intentional or incidental – is arguably slow, effortful and constrained by self-protective processes.

The consensus process also revealed clear gaps in the self-knowledge literature that demand attention: theoretical frameworks remain underspecified, measurement practices often lack comparability, and the mechanisms and boundary conditions of self-knowledge are poorly understood. Addressing these challenges will require conceptual elaboration through the development of more integrative theoretical frameworks and shared taxonomies as well as coordinated empirical efforts, including adversarial collaborations^{125,126}.

There are several limitations to consensus processes in general and our process in particular. First, consensus does not imply correctness: it reflects collective expert judgement rather than empirical validation, and different expert panels might reach somewhat different conclusions depending on their expertise and disciplinary focus. All panel members in our consensus process were affiliated with institutions in Western countries, and disciplinary representation was concentrated in personality and social psychology. Consequently, perspectives from non-Western cultural contexts and from other psychological subdisciplines and methodological traditions were under-represented. Thus, the conclusions might not fully capture cultural and interdisciplinary variability, underscoring the need for future work to adopt more global and diverse perspectives. Second, although iterative feedback across rounds fosters convergence, it might also narrow the

diversity of viewpoints. Relatedly, although the facilitators sought to remain neutral, the phrasing of statements and summaries by the facilitators might have subtly shaped the trajectory of agreement. Taken together, these considerations highlight that the present findings should be viewed as a systematic synthesis of expert perspectives rather than definitive empirical conclusions, and as a foundation for future theoretical and empirical work.

In conclusion, the insights from our consensus process (summarized in Fig. 3) suggest that research on self-knowledge must embrace nuance, not only asking what self-knowledge is but how it functions, when it matters and for whom. By systematically unpacking its antecedents, mechanisms and consequences, and by incorporating perspectives from diverse cultural and disciplinary contexts, future research can shed even more light on one of psychology’s most enduring questions – how well people know themselves – and what that means for their experience, relationships and personal growth.

Published online: 8 April 2026

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Acknowledgements

The authors thank D. Leising for his invaluable advice and guidance throughout the process. This research was funded by a grant from the European Union awarded to the first author (KNOW-THYSELF; project no. 101039433). The views and opinions expressed are those of the authors only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency.

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I.T., M.B., N.C. and A.S. researched data for the article. I.T., M.D.B., W.B., H.B., E.N.C., M.D., G.H., W.H., C.J.H., L.J.H., C.H.J., J.I.K., J.L., R.R., L.D.S., N.S., J.S. and S.V. contributed substantially to discussion of the content. I.T. wrote the article. All authors reviewed and/or edited the manuscript before submission.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s44159-026-00554-1>.

Peer review information *Nature Reviews Psychology* thanks Hu Chuan-Peng, Rebecca Schlegel and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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Consensus statement

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